



Signal or luck?

Measuring what drives citation in AI search engines

When ChatGPT, Perplexity, or Claude cites a page, is that a real signal about the page, or mostly chance?

Mehreen | Capstone proposal

Supervised by: Professor George Hagstrom

Same question, three runs, three source lists

- AI engines answer directly and cite only a few sources
- Being cited is the new being ranked, but there is no Search Console for it
- In my preliminary data, only 14% of retrieved URLs were stable across 3 runs (Perplexity, 40 prompts)

“Best email marketing tools for small business” (Perplexity)

Run 1 · top pages	Run 2 · top pages	Run 3 · top pages
zoho.com	mailersend.com	ventureharbour.com
brevo.com	pcmag.com	mailersend.com
capterra.com	ventureharbour.com	emailtooltester.com
business.com	guardian.ng	learn.g2.com
pcmag.com	emailtooltester.com	pcmag.com

Teal = retrieved in all three runs.

Who cares: publishers and marketers investing in GEO, and researchers asking whether AI search makes information access less stable.

Prior work

What we know

- Only 34–42% overlap in cited sources between repeated runs (Schulte et al., 2026)
- Content tricks mostly fail; what matters most is where a page sits in the results the AI reads (Puerto et al., 2025)
- No technique reliably gets a page discovered (Martinez, 2026)

What's missing

- How much citation is chance versus content versus rank, in one design
- Whether a source's chance of being cited depends on the competing sources
- A repeatable way to measure visibility across engines

Three tests: noise, context, signal

1 Noise

Same question again: how much do the cited sources change?

Run each prompt 5–8 times per engine; measure overlap with confidence intervals (Jaccard, RBO)

2 Context

Add or remove one source: do the others' odds of citation stay in proportion?

Coffee vs tea 2:1 shouldn't change when juice joins the menu. Test on a controlled document set (choice model, Plackett-Luce)

3 Signal

Beyond search rank, do length, statistics, or wording predict citation?

Logistic regression on controlled documents, then on real Google results (correlational)

Two datasets in hand, one to build

AI citations

1,728

citations across 898 domains, 3 engines, 100 prompts

COLLECTED

Google comparison

765 pages

Top-10 results per prompt: ranked on Google, cited or not

COLLECTED

Controlled documents

Selected Pages

Known features; shown to models via chatlas / ellmer

TO BUILD, WEEKS 1–4

Preliminary result, still to validate: 14% of retrieved URLs stable across 3 Perplexity runs (40 prompts). Perplexity's retrieved and cited lists may overlap heavily, so I will confirm they are distinct signals before calling this a finding.

12 weeks, clear checks, and a plan for cuts



Midterm check

Test 1 done for all three engines with confidence intervals and one summary figure; Test 2 pipeline validated.

Final check

IIA tested on at least two engines across many compositions; Test 3 compared with and without content features. Null results count as findings.

Scope, Cut order: boosting model, then the Google half of Test 3, then the third engine in Test 2.

Main risk: Raw responses saved to guard against API changes.

Cost: API fees only.